

RS004 Week 13 — English Worksheet

Topic: Enemies & Game Over · **Course:** Platformer Game · **Time:** about 45 minutes

This week your game gets **enemies**. You build an **Enemy** class that patrols left and right on its own. Touching an enemy triggers **Game Over**, and pressing **R** restarts. A **game_state** variable decides what the game is doing.

Keep these words handy: **class**, **patrol**, **game_state**, **Game Over**, **restart (R)**.

1 · Predict 🧠

Read each snippet. Before you run it, write what you think will happen.

```
def update(self):
    self.rect.x = self.rect.x + self.direction * self.speed
    if self.rect.x > self.start_x + self.range_x:
        self.direction = -1
    if self.rect.x < self.start_x:
        self.direction = 1
```

The enemy reaches the right end of its range. What happens to **direction** ?


```
if player.colliderect(enemy.rect):
    game_state = "gameover"
```

The player touches an enemy. What does the game switch to?


```
elif game_state == "gameover":  
    if keys[pygame.K_r]:  
        player.x = start_x  
        player.y = start_y  
        game_state = "playing"
```

The game is over and R is pressed. What resets, and what does the game become?

2 · Read the Class

An `Enemy` is created with `Enemy(300, 530, 40, 40, 200)` . Match each value.

1. 300 2. 530 3. 40 4. 40 5. 200

A. patrol range (how far it walks)
B. start x
C. start y
D. width
E. height

1 → __ 2 → __ 3 → __ 4 → __ 5 → __

3 · Spot the Bug 🐛

Each snippet is broken. Rewrite it so it works, then explain why the original was wrong.

Bug A — The enemy should walk back and forth. Right now it walks off the right side and never comes back.

```
def update(self):  
    self.rect.x = self.rect.x + self.speed
```

Hint: there is no `direction`, so it only ever moves one way. It needs to flip at the edges.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Movement should stop once the game is over. Right now the player keeps moving on the Game Over screen.

```
keys = pygame.key.get_pressed()  
if keys[pygame.K_LEFT]:  
    player.x -= PLAYER_SPEED  
if keys[pygame.K_RIGHT]:  
    player.x += PLAYER_SPEED  
for enemy in enemies:  
    enemy.update()  
    if player.colliderect(enemy.rect):  
        game_state = "gameover"
```

Hint: wrap the play logic so it only runs when `game_state == "playing"`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — R should restart the game, but pressing it does nothing.

```
if game_state == "gameover":  
    player.x = start_x  
    player.y = start_y  
    game_state = "playing"
```

Hint: this restarts every frame with no key check. Only restart when R is pressed.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Populate & Polish

Start from your working enemies.

1. Add a second enemy with a *different* patrol range. Write its `Enemy(...)` line.

2. Change an enemy's speed. Does a faster enemy make the level harder or just annoying?

3. **Stretch:** show the final score on the Game Over screen. Where do you draw it?

5 · Build & Show 📷

Add patrolling enemies to your own level. Place at least three, and make sure the level is still clearable.

When it works, send a **photo or video** showing you dodging enemies *and* a Game Over, then explain what you did. Use these starters — write 4 to 6 sentences.

First, my **Enemy** class patrols by ...

The **direction** flips when ...

Touching an enemy sets `game_state` to ...

I restart with R by ...

One tricky moment was ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video showing dodging, dying, and restarting. Teach it to someone new. Try to use these words: **class**, **patrol**, **game_state**, **Game Over**, **restart**.

1. Show an enemy patrolling back and forth.
2. Dodge one, then touch one and trigger Game Over.
3. Read the patrol **update** out loud and explain the direction flip.
4. Press R and show the restart.

Write what you will say in your video. Plan it here first.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.